An Interactive Learning and Research Platform for Real-Time Adaptive Feedback Cancellation Systems

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Abstract. Acoustic feedback is a persistent challenge in audio closed-loop systems such as hearing aids and public address systems, occurring when the sound emitted from the loudspeaker is recaptured by a microphone. The acoustic feedback phenomenon limits stable gain and causes howling under some circumstances. A common and effective solution is adaptive feedback cancellation (AFC), which uses adaptive filters to cancel the feedback signal, enhancing stability. A key difficulty for AFC is the inherent correlation between the loudspeaker signal and the desired input (especially when the input signals are spectrally colored, like speech, music, etc.), which leads to biased estimates of the feedback path. To counter this, methodologies such as the prediction error method (PEM) and its variant decorrelate the input signals of the adaptive filter, improving estimation robustness. The PEM integrated with a range of adaptive algorithms, including NLMS, IPNLMS, H-NLMS, APA, and IPAPA, has been proposed in the literature, each balancing convergence speed, computation, and steady-state error. Since implementing and comparing these algorithms in a unified, real-time context is still challenging, we introduce a comprehensive MATLAB/Simulink platform that provides an end-to-end AFC system. This platform enables users to seamlessly switch between various AFC algorithms, toggle pre-filtering, and monitor key performance metrics, such as normalized misalignment (MIS) and added stable gain (ASG) in real-time. This interactive environment serves as both a powerful educational tool and an efficient testbed for validating novel AFC techniques.

Keywords: Acoustic feedback cancellation, PEM, real-time AFC platform, interactive learning and research.

1 Introduction

In modern audio applications, the problem of acoustic feedback remains a critical obstacle in closed-loop systems such as hearing aids and public address systems [1]. This phenomenon arises from the acoustic coupling between a system's loudspeaker and microphone, creating an unstable signal loop that leads to audible howling artifacts, severe degradation of sound quality, and limits maximum achievable stable gain [2]. The most prevalent solution is adaptive feedback cancellation (AFC), which employs an adaptive filter to model the acoustic feedback path. By generating an accurate estimate of this

path, the system can compute the feedback signal and subtract it from the microphone signal, thereby canceling the feedback signal and restoring stability [1], [2], [3]. A primary challenge in AFC methods is the inherent correlation between the loudspeaker signal and the incoming signal, which introduces a bias in the adaptive filter's estimation process, particularly with spectrally colored inputs like speech and music [4], [5]. To mitigate this issue, decorrelation techniques are essential. The prediction error method (PEM) is a widely adopted approach that pre-whitens the signals by modeling the input source. This process minimizes the statistical dependence between the signals, enabling a more accurate and unbiased estimation of the feedback path [6], [7].

To implement the adaptive filter, a diverse range of algorithms has been proposed. The foundational algorithm is the normalized least mean squares (NLMS), which is simple but yields slow convergence and tracking rate, especially when the incoming signals are spectrally colored. To address the limitations of the foundational method, other AFC algorithms have been developed, including the fast-converging algorithms like the affine projection algorithm (APA) and its variants [8], [9]. In the methods based on APA, the most recent P input vectors are used, where P, the projection order, defines how many past inputs inform the adaptive filter’s coefficient estimation.

Recently, Acoustic Feedback Cancellation (AFC) has seen a surge in interest in approaches based on proportionate-type algorithms. These methods leverage the sparse nature of the feedback path by assigning variable weights to the adaptive filter coefficients based on their magnitude. This allows larger coefficients to be updated more aggressively, leading to enhanced convergence rate, better tracking ability, and improved sound quality. Some notable sparse adaptive filtering techniques introduced are the (Improved) Proportionate Normalized Least Mean Squares, or (I)PNLMS, and its variants [10], [11], [12], [13], [14]. Moreover, researchers have explored combining proportionate-type algorithms with fast-adaptive algorithms (like the APA and the affine projection sign algorithm (APSA)) [2], [15], [16], [17], and robust subband algorithms for sparse/block-sparse system identification [18].

Other advanced methods have been investigated; for instance, the Kalman filter has been explored for optimal state estimation, such as its integration in the frequency domain with the PEM to enhance performance [19]. To improve robustness against abrupt changes in the feedback path, lightweight state-detection models have been combined with the Kalman filter to enable rapid recovery [20].

MATLAB and Simulink are the de facto standards for modeling audio systems. Existing research has predominantly focused on offline simulations of standalone AFC algorithms in MATLAB, thereby limiting the ability to visualize various AFC algorithms in real time. To address this limitation, this work introduces an end-to-end AFC platform in Simulink that enables users to experiment with multiple adaptive algorithms, analyze pre-filtering effects, and visualize key performance metrics in real-time. It also provides a foundational testbed that bridges theory and practice, serving as a

valuable tool for both education and research in the field. To the best of our knowledge, no prior work has investigated a comprehensive AFC platform built on a Simulink environment, enabling interactive, real-time simulation of different adaptive feedback cancellation (AFC) algorithms.

2 Design and Implementation of the proposed AFC Platform

2.1 Overall System Architecture in Simulink

The overall architecture of the proposed AFC platform is designed and implemented entirely within the Simulink environment in MATLAB, see Figure 1. This end-to-end platform comprises several signal processing blocks interconnected to simulate a real-time AFC system, such as the main AFC processing subsystem, the Input signal source, and the AFC_Performance_Monitor. By leveraging standard DSP System Toolbox blocks and employing frame-based processing, the proposed platform achieves the low latency required for real-time simulations without relying on external third-party libraries.

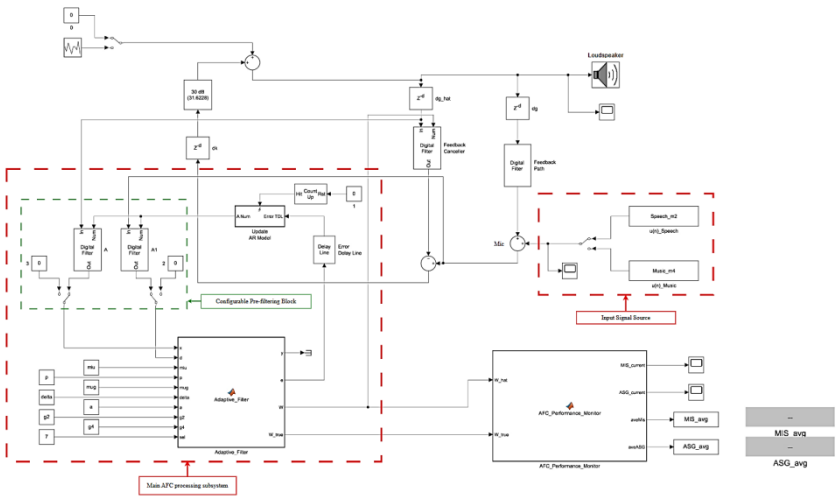


Fig. 1. Overall architecture of the proposed AFC platform in Simulink.

The block of the Input signal source provides flexibility to select among different audio types of the desired signals, such as speech and/or music, aiming at simulating diverse operating conditions. Both the incoming signal and the feedback signal go through the Mic block—representing a microphone—and combine into a microphone signal. That composite signal is then sent to the main AFC processing subsystem. The feedback path, implemented via a digital filter, emulates how sound would propagate from the loudspeaker back to the microphone in a real physical environment.

The main AFC processing subsystem is the core component of the entire platform, featuring an adaptive filter block where various algorithms like NLMS, IPNLMS l_1-norm, APA, and IPAPA are implemented and can be easily selected by the user. Furthermore, this subsystem contains a configurable pre-filtering block that can be enabled or disabled to evaluate its contribution to the signal decorrelation. Finally, to provide clear visual results, the platform integrates a dedicated performance monitoring block (called AFC performance monitor) that allows the computation of evaluation metrics in real-time. We also use scope blocks to display the resulting normalized misalignment (MIS), added stable gain (ASG) graphs, as well as to monitor the microphone and loudspeaker signals. The entire architecture is designed to be modular, allowing users to easily interact, modify parameters, and evaluate the performance of each component within the system.

3 Core Signal Processing Blocks

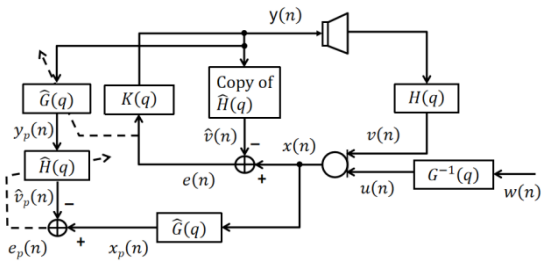


Fig. 2. Block diagram of the proposed AFC system.

To provide a clearer understanding of the internal signal flow and the interactions among the core processing modules, a block diagram of the proposed AFC system is illustrated in Figure 2. While Figure 1 presents the overall Simulink-based architecture of the end-to-end AFC platform, Figure 2 focuses on the essential signal processing mechanisms that form the functional backbone of the system.

This structure is an implementation of the PEM, a technique designed to reduce bias in the feedback path estimate. The method operates by modeling the incoming signal via an all-pole filter $G^{-1}(q)$ and then using an estimated inverse model $\hat{G}(q)$ to pre-whiten the inputs of the adaptive filter $\hat{H}(q)$. This interaction between the pre-filtering path, the adaptive filtering stage, and the forward path $K(q)$ allows for a more accurate estimation and effective cancellation of the acoustic feedback. The following subsections describe each of these processing blocks in detail.

Pre-filtering Block

A main challenge in AFC systems is the correlation between the incoming signal (e.g., speech) and the loudspeaker signal, which arises from the closed-loop nature of the system itself [21]. This correlation induces a bias in the impulse response estimation

process of the adaptive filter, which degrades accuracy and can potentially lead to system instability. To address this issue, the platform incorporates a configurable pre-filtering block [7].

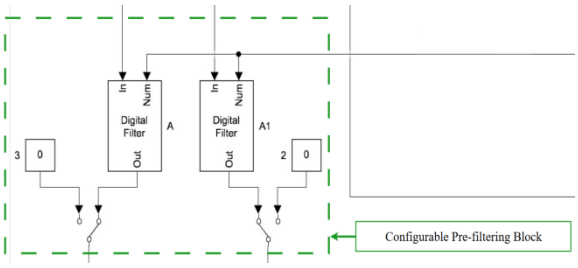


Fig. 3. Configurable pre-filtering block of the proposed AFC platform in Simulink.

The primary role of this block is to decorrelate, or "pre-whiten" the signals before they are fed into the adaptive filter. As illustrated in Figure 3, this block consists of two digital filters (labeled A and A1) that operate in parallel to process the loudspeaker and microphone signals, respectively. These filters are based on an estimated model of the incoming signal, typically an auto-regressive (AR) model for speech. By passing the loudspeaker and microphone signals through the inverse of this estimated incoming signal model ($\hat{G}(q)$), their statistical dependence is minimized, allowing the adaptive filter to converge to a less biased and more accurate solution [7].

To demonstrate the pre-filter's impact, the platform features a real-time switch that allows users to enable or bypass it. This provides a direct way to evaluate how pre-whitening the incoming signals reduces estimation bias and enhances system performance. When the pre-filter is bypassed, the still-correlated signals cause the adaptive filter to exhibit degraded performance, such as slower convergence and higher steady-state MIS. By comparing the two modes, users can quantitatively assess the performance improvement gained from decorrelation, thereby reinforcing their understanding of the trade-offs between system complexity and accuracy.

Adaptive Filter Block

The adaptive filter block serves as the central processing of the learning platform, implemented as a MATLAB function block within Simulink. This block is designed to store several AFC algorithms from common algorithms like NLMS and APA to state-of-the-art algorithms such as IPNLMS, HNLMS, and IPAPA, and the user can dynamically switch among them in real-time via a single input parameter, “sel”.

A key highlight and the core value of this block is its high extensibility, a critical feature for both research and educational purposes. The internal structure of the MATLAB function code is explicitly organized using an if-elseif conditional chain, where each branch corresponds to a specific algorithm. This design allows researchers and students to seamlessly integrate new or custom algorithms into the platform. The extension process can be accomplished simply by adding a new elseif branch to the

existing source code containing the weight-update logic for the new algorithm. Alternatively, the modular nature of Simulink allows a user to replace this function block entirely with a new one, leaving other system components unaffected, provided the input/output interface is maintained. This flexible extensibility transforms the platform from a mere demonstration tool into a powerful testbed for rapid prototyping, benchmarking, and validating novel feedback cancellation methods.

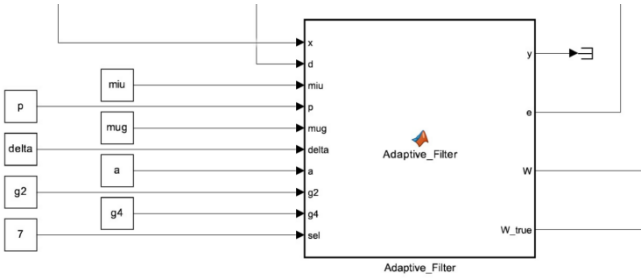


Fig. 4. Adaptive filter block of the proposed AFC platform in Simulink.

To illustrate the functionality of the various adaptive algorithms contained within the adaptive filter block, we provide a brief review of those algorithms below.

The PEM-NLMS method, where NLMS serves as an adaptive algorithm providing a baseline for evaluating more advanced methods. Here, the NLMS algorithm is employed as a fundamental adaptive filtering technique for estimating the impulse response (IR) of the acoustic feedback path. In the system model, see Figure 2, the microphone signal $x(n)$ is defined as a summation of the incoming signal $u(n)$ and the feedback signal $v(n) = \boldsymbol{h}^T(n)\boldsymbol{y}(n)$, where $\boldsymbol{h}(n)$ is the L_h-dimensional IR vector of the true feedback path and $\mathbf{y}(\mathrm{n})$ is the L_h-dimensional vector of the loudspeaker signal. The adaptive filter is responsible for generating an estimated feedback path, while the error signal $e(n)$ quantifies the deviation between the actual microphone signal and the estimated feedback signal, formulated as:

$$e(n) = x(n) - \hat{\boldsymbol{h}}^T(n)\boldsymbol{y}(n). \tag{1}$$

To enhance estimation accuracy, particularly when the incoming signal $u(\mathrm{n})$ exhibits spectral coloration, a pre-whitening stage is applied to the adaptive filter inputs. Assuming that the forward path $K(q)$ includes a delay d_k and an amplifier with broadband gain $|K|$, defined as $K(q) = |K|\, q^{-d_k}$ with q^{-1} denoting the delay operator [11]. The $x(n)$ and $y(n)$ are processed through an estimated filter $\hat{G}(q)$ which is an inverse of the incoming signal model. This produces the pre-whitened signals $x_p(n) = \hat{G}(q)x(n)$ and $y_p(n) = \hat{G}(q)y(n)$, leading to the expression for the pre-whitened error signal as

$$e_p(n) = x_p(n) - \hat{\boldsymbol{h}}^T(n)\boldsymbol{y}_p(n),$$

where $\boldsymbol{y}_p(n) = [y_p(n), y_p(n-1), \dots, y_p(n - L_{\hat{h}} + 1)]^T$. (2)

In the PEM-NLMS, adaptive filter coefficients $\widehat{\boldsymbol{h}}(n)$ are updated iteratively using NLMS algorithm as follows

$$\widehat{\boldsymbol{h}}(n) = \widehat{\boldsymbol{h}}(n-1) + \frac{\mu}{(\|\boldsymbol{y}_p(n)\|_2^2+\delta)}\boldsymbol{y}_p(n)e_p(n). \tag{3}$$

Here, μ denotes the step-size controlling the convergence rate and stability of adaptation, while δ is a small regularization parameter introduced to prevent division by zero. By incorporating this pre-whitening stage and normalization, the PEM-NLMS framework effectively mitigates the estimation bias found in conventional LMS-based methods, achieving improved convergence and robust tracking performance for real-time acoustic feedback control [11].

To overcome the convergence speed limitations of NLMS for highly correlated signals, the APA extends its core concept by utilizing a set of recent input vectors to perform a more robust and informed coefficient update. While NLMS uses only the most recent input vector, APA generalizes this approach by incorporating a set of the most recent P input vectors, with P known as the projection order. This mechanism allows the algorithm to project the desired response onto a subspace spanned by these vectors, enabling faster convergence and better numerical stability in correlated environments [9], [22]. Mathematically, the iterative coefficient update of APA is expressed as

$$\widehat{\boldsymbol{h}}(n) = \widehat{\boldsymbol{h}}\,(n-1) + \mu\boldsymbol{Y}(n)[\delta\boldsymbol{I}_P + \boldsymbol{Y}^T(n)\boldsymbol{Y}(n)]^{-1}\boldsymbol{e}_{APA}(n), \tag{4}$$

where the matrix $\boldsymbol{Y}(n)$ is constructed from the P most recent input vectors as $\boldsymbol{Y}(n) = [\boldsymbol{y}(n), \boldsymbol{y}(n-1), \dots, \boldsymbol{y}(n-P+1)]$. The desired signal samples are represented by the vector $\boldsymbol{x}_{APA}(n) = [x(n), x(n-1), \dots, x(n-P+1)]^T$ and the error signal vector, which drives the adaptation, is given by

$$\boldsymbol{e}_{APA}(n) = \boldsymbol{x}_{APA}(n) - \boldsymbol{Y}^T(n)\widehat{\boldsymbol{h}}(n-1). \tag{5}$$

By exploiting spatial information of multiple recent input vectors, the APA yields a superior convergence rate and higher estimation accuracy. This enhanced performance, however, comes at the cost of significantly increased computational complexity due to the matrix operations required at each iteration.

While the APA enhances convergence by leveraging the temporal history of the adaptive filter's input signal, an alternative class of algorithms improves performance by exploiting the structural properties of the feedback path itself. This is the principle behind proportionate adaptation, exemplified by IPNLMS, which is specifically designed to excel when the impulse response is sparse.

Building upon the NLMS framework, the IPNLMS algorithm introduces an adaptive weighting mechanism that accelerates convergence by proportionally adjusting the update rate of each filter coefficient. This method is particularly effective in scenarios where the acoustic feedback path exhibits sparsity. The IPNLMS with the l_1-norm formulation refines the original proportionate NLMS by incorporating a smoother transition between proportionate and uniform adaptation modes, thereby improving the balance between convergence speed and steady-state error [11].

Like the NLMS case, the pre-whitened signals $\boldsymbol{y}_p(n)$ and the pre-whitened error signal $e_p(n)$ are obtained from the PEM process. The difference is in computing the adaptive filter coefficients $\widehat{\boldsymbol{h}}(n)$ iteratively as

$$\widehat{\boldsymbol{h}}(n) = \widehat{\boldsymbol{h}}\,(n-1) + \frac{\mu \boldsymbol{A}(n-1)}{\boldsymbol{y}_p^T(n)\boldsymbol{A}(n-1)\boldsymbol{y}_p(n) + \delta}\boldsymbol{y}_p(n)e_p(n), \tag{6}$$

where $\boldsymbol{A}(n-1)$ is a diagonal weighting matrix defined as

$$\boldsymbol{A}(n-1) = diag\left\{a_0(n-1), \dots, a_{L_{\widehat{h}}-1}(n-1)\right\}. \tag{7}$$

Each element $a_l(n)$ determines the local adaptation rate for the corresponding coefficient $\hat{h}_l(n)$. The IPNLMS algorithm based on the l_1-norm defines the proportional weighting factor $\lambda_l(n)$ as

$$\lambda_l(n) = (1-\alpha)\frac{||\widehat{\boldsymbol{h}}(n)||_{l_1}}{L_{\widehat{h}}} + (1+\alpha)\left|\hat{h}_l(n)\right|, \tag{8}$$

where $-1 \leq \alpha < 1$ is a control parameter balancing the proportionate and uniform adaptation. Using $\lambda_l(n)$, the normalized weighting coefficients $a_l(n)$ are derived as

$$a_l(n) = \frac{1-\alpha}{2L_{\widehat{h}}} + (1+\alpha)\frac{|\hat{h}_l(n)|}{2||\widehat{\boldsymbol{h}}(n)||_{l_1} + \varepsilon}, \tag{9}$$

where ε is a small positive constant that prevents numerical instability. When $\alpha = -1$, the IPNLMS reduces to the standard NLMS algorithm, as the step-size becomes uniformly distributed among all filter taps. In contrast, when α approaches 1, the IPNLMS behaves similarly to the original proportionate NLMS, assigning larger updates to coefficients with higher magnitudes. Practical implementations often use values of α between -0.5 and 0 to achieve a compromise between convergence rate and steady-state accuracy.

A natural evolution is to combine the proportionate mechanism of IPNLMS and the multi-vector projection framework of APA, called IPAPA [2]. Building upon the principles of IPNLMS, IPAPA extends the idea of proportionate adaptation to the affine projection framework. Its key advantage lies in dynamically distributing the adaptation gain among the filter coefficients based on their relative magnitudes, enabling faster tracking of dominant coefficients and improved performance for sparse feedback paths. Mathematically, the coefficient update equation of IPAPA is defined as follows

$$\widehat{\boldsymbol{h}}(n) = \widehat{\boldsymbol{h}}(n-1) + \mu \boldsymbol{Y}(n)\boldsymbol{A}(n-1)[\delta \boldsymbol{I}_P + \boldsymbol{Y}^T(n)\boldsymbol{A}(n-1)\boldsymbol{Y}(n)]^{-1}\boldsymbol{e}_{APA}(n). \tag{10}$$

While IPAPA enhances both convergence speed and estimation accuracy, it does not explicitly address the issue of system instability. To tackle this challenge, an alternative approach focuses on ensuring robust and fail-safe operation. This is the principle behind the H-NLMS algorithm [23], which combines the high accuracy of PEM-NLMS with the robust stability of the conventional NLMS algorithm. The H-NLMS functions as a switched system that dynamically alternates between these two modes based on system stability. This selection is governed by a binary control signal $\kappa(n)$,

$$\kappa(n) = \begin{cases} 1, if \left| e_p(n) - e_L(n) \right| > c \\ 0, \qquad\qquad otherwise \end{cases}, \tag{11}$$

which is determined by comparing the difference between the $e_L(n) = \beta . tanh(\frac{e_p(n)}{\beta})$ and the $e_p(n)$, with β a scaling parameter determining the range of the soft-clipping nonlinearity, and c a decision threshold governing the sensitivity of the switching mechanism. The update rule of the adaptive filter is given by

$$\hat{\boldsymbol{h}}(n) = \hat{\boldsymbol{h}}(n-1) + [1 - \kappa(n)]\mu_1(n)\tilde{\boldsymbol{y}}(n)\tilde{e}_L(n) + \kappa(n)\mu_2(n)\boldsymbol{y}(n)e_L(n), \tag{12}$$

with $\tilde{e}_L(n) = \tilde{x}(n) - \hat{\boldsymbol{h}}^T(n)\tilde{\boldsymbol{y}}(n)$. Here, $\tilde{x}(n)$ and $\tilde{\boldsymbol{y}}(n)$ are defined as $\tilde{x}(n) = \hat{G}(q)x(n)$ and $\tilde{\boldsymbol{y}}(n) = \hat{G}(q)\boldsymbol{y}(n)$. The algorithm utilizes two distinct step sizes, a small step size $\mu_1(n) = \frac{\mu_1}{||\tilde{\boldsymbol{y}}(n)||^2 + \delta}$ is used for the PEM-NLMS mode when $\kappa(n) = 0$, ensuring low steady-state error. When instability or howling is detected, $\kappa(n) = 1$, the system instantly switches to the NLMS mode with a larger step size $\mu_2(n) = \frac{\mu_2}{||\boldsymbol{y}(n)||^2 + \delta}$ to ensure rapid recovery.

The key advantage of H-NLMS is this dynamic switching mechanism. Unlike algorithms such as IPNLMS or IPAPA that focus on accelerating convergence speed for sparse systems, H-NLMS is specifically designed to achieve a unique trade-off; it maintains the high accuracy of PEM-NLMS during stable operation while providing the fail-safe robustness of standard NLMS to suppress instability, resulting in a system that is both precise and highly reliable.

4 Monitor normalized misalignment (MIS) and added stable gain (ASG) in real-time

To evaluate the performance of AFC algorithms, two primary metrics are commonly used: normalized misalignment and added stable gain.

4.1 Normalized misalignment (MIS)

MIS serves as a crucial metric for measuring the accuracy of an adaptive filter's estimation of the acoustic feedback path. It quantifies the difference between the adaptive filter's coefficients and the true, unknown coefficients of the feedback path [7].

$$\text{MIS} = 10log_{10}(\frac{||\boldsymbol{h} - \hat{\boldsymbol{h}}||_2^2}{||\boldsymbol{h}||_2^2}). \tag{13}$$

A key objective in optimizing the AFC algorithm performance is to reduce this misalignment as much as possible. A lower MIS value indicates that the adaptive filter is more accurately modeling the unknown system, which contributes to overall system effectiveness.

4.2 Added Stable Gain (ASG)

ASG is another key performance measure that evaluates the improvement in system stability resulting from the implementation of a feedback cancellation algorithm [24].

$$\mathrm{ASG} = 10log_{10}\frac{1}{max_{\Omega}|H(\Omega)-\hat{H}(\Omega)|^2} - 10log_{10}\frac{1}{max_{\Omega}|H(\Omega)|^2}. \quad (14)$$

In the context of hearing aid devices (HADs), the ability to increase the available gain is a critical parameter, as it determines whether the device can effectively compensate for the user's hearing loss. A high ASG indicates that the system can handle significantly greater amplification before howling occurs, thereby substantially improving the user experience.

For immediate visual feedback on system performance, visualization tools like Simulink scopes and display blocks can be employed to show performance metrics such as MIS and ASG in real time. This allows for an easy observation of the algorithm's convergence and an assessment of its impact on system stability.

5 Experimental Results and Comparative Analysis

5.1 Simulation Setup

To evaluate the performance of the adaptive algorithms, a realistic and challenging test environment was established. The feedback paths used in the simulation were based on sparse impulse responses, derived from real-world scenarios such as free-field and telephone-near conditions [25].

A standardized set of incoming signals was meticulously prepared to create an authentic testing environment. The input stimuli focused on a primary acoustic scenario: structured, harmonic signals (speech). The source materials included speech signals from the NOIZEUS database (comprising both male and female utterances) [26]. Instead of using these digital sources directly, a meticulous recording procedure was implemented to capture the signals under simulated real-world listening conditions. The source signals were reproduced in an anechoic environment and recorded by a microphone placed in the right ear of a Head and Torso Simulator (HATS). To introduce spatial variability, recordings were conducted for two separate source positions: one directly in front of the HATS and another positioned to its right. This protocol yielded a total of two distinct audio files for testing. All signals were captured at a sampling frequency of 16 kHz and were peak-normalized to ensure a maximum absolute amplitude of 1.0 [25]. This final step establishes a consistent energy baseline, enabling a fair and repeatable comparative analysis across all adaptive feedback cancellation algorithms in the platform. To quantitatively evaluate the tracking capability and stability of the AFC algorithms under these test conditions, the platform records two key performance metrics, MIS and ASG.

To ensure a consistent and fair comparison among all the evaluated algorithms, the adaptive filter parameters were uniformly configured. The filter length was fixed at $L = 22$ for all cases. For the NLMS, IPNLMS l_1-norm, APA, and IPAPA algorithms,

the step size parameter was set to $\mu = 0.001$ while the projection order was defined as $P = 2$. The delay in the forward path was $d_k = 32$ samples, the delay of the feedback cancellation path was fixed at $d_g = 16$ samples, and $|K| = 30\ dB$. In the case of the H-NLMS algorithm, two separate step sizes were used, $\mu_1 = 0.001$ for the PEM-NLMS and $\mu_2 = 0.2$ for the NLMS. The soft-clipping parameters were chosen as $\beta = 2$, $c = 0.15$, and other parameters were selected as $\delta = 10^{-6}$, $\alpha = 0$.

5.2 Performance Analysis

The performance of five AFC algorithms—NLMS, IPNLMS l_1-norm, APA, IPAPA, and H-NLMS—was evaluated within the PEM framework using the proposed platform. The analysis was conducted over a 60-second duration using a recorded speech signal, with a sudden change in the feedback path after 30 seconds to test tracking capabilities. The performance was quantified by the MIS over time, as depicted in Figures 5 and 6, and by the average MIS ($\overline{MIS}$) and average ASG ($\overline{ASG}$) as summarized in Table 1.

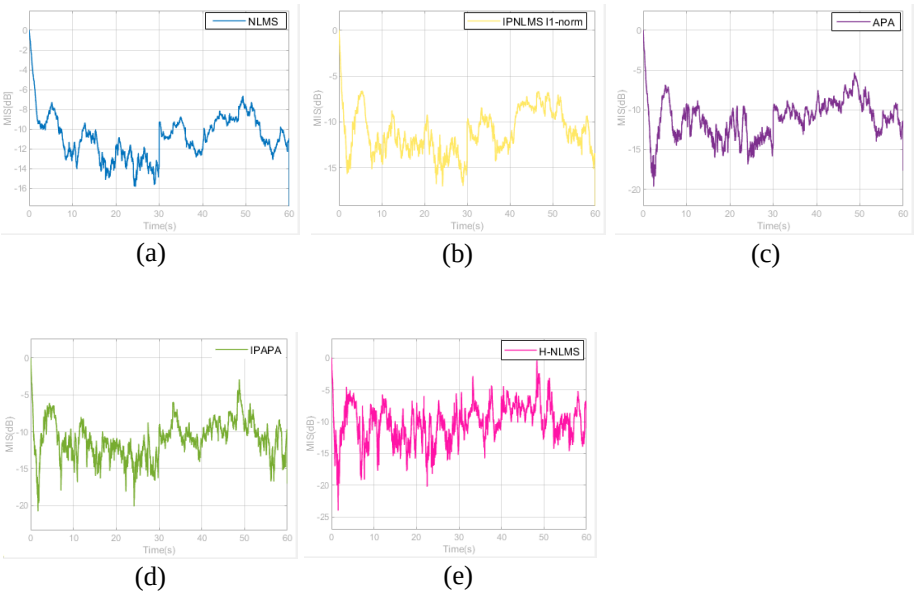


Fig. 5. MIS of the PEM using (a) NLMS, (b) IPNLMS l_1-norm, (c) APA, (d) IPAPA, (e) H-NLMS algorithms with speech incoming signal.

The classic NLMS algorithm, used as a baseline, demonstrates its known limitations with a slow initial convergence rate, as shown by its gradual initial descent in Figure 5a. In contrast, advanced algorithms like APA and IPAPA showcase their primary advantage a vastly superior convergence speed. This is visually evident in the sharp, vertical drops in MIS for APA (Figure 5c) and IPAPA (Figure 5d) within the first few seconds, allowing them to adapt to the acoustic environment almost instantly. The

IPNLMS algorithm (Figure 5b) also shows a significantly faster convergence than the baseline NLMS.

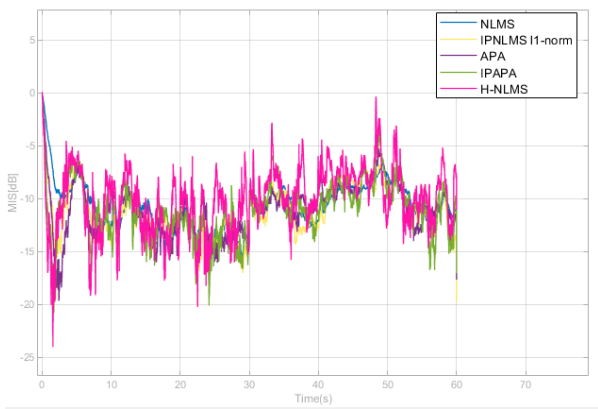


Fig. 6. Compares MIS of the PEM using different adaptive algorithms with speech incoming signal.

This rapid adaptation is a critical performance gain. While the advanced algorithms exhibit higher steady-state fluctuations, visible across Figures 5b, 5c, and 5d, this is a justified trade-off for their ability to quickly track the change of the feedback path. This superior tracking is clearly demonstrated by their swift recovery after the 30-second mark in the comparative overview of Figure 6. The quantitative results in Table 1 match well with the observation in Figure 6, with IPNLMS and IPAPA demonstrating excellent average performance, highlighting the superiority of proportionate and affine projection techniques. The H-NLMS algorithm represents a different design trade-off, prioritizing robust stability over optimal steady-state accuracy, which explains the highly volatile MIS curve seen in Figure 5e.

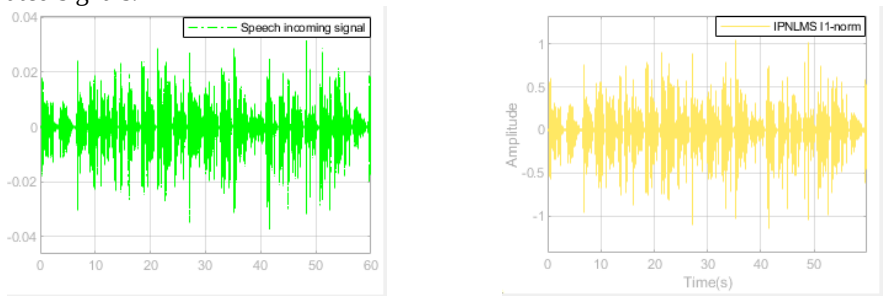
Table 1. Average misalignment and average added stable gain for the PEM using NLMS, IPNLMS l_1-norm, APA, IPAPA, H-NLMS algorithms with the speech incoming signal.

AFC methods	$\overline{MIS}$	$\overline{\text{ASG}}$
NLMS	-10.7590	11.3584
IPNLMS l_1-norm	-11.1534	11.6041
APA	-11.1121	11.5744
IPAPA	-11.0543	11.3800
H-NLMS	-10.0723	10.6803

While the Figures show the convergence dynamics, Table 1 provides a quantitative summary of the average performance. The IPNLMS l_1-norm algorithm achieved the best average MIS (-11.1534 dB) and the highest average ASG (11.6041 dB). This result

validates its strength in achieving high steady-state accuracy, a benefit of its proportionate update mechanism, which is particularly effective for the sparse nature of acoustic feedback paths. The APA and IPAPA also show strong average performance, significantly outperforming the NLMS baseline and confirming the value of the affine projection and proportionate adaptation frameworks.

Conversely, the H-NLMS algorithm recorded the highest average MIS (-10.0723 dB) and lowest average ASG (10.6803 dB). This outcome is not a failure but a direct reflection of its specific design objective. The primary goal of H-NLMS is not to achieve the lowest possible steady-state error, but to ensure robust stability and rapid recovery from howling. The trade-off for this enhanced reliability is a compromise in average estimation accuracy compared to algorithms purely optimized for performance. This highlights an important trade-off: sacrificing some steady state precision for guaranteed stability. The results for the conventional NLMS serve as a baseline, confirming its well-known limitations in convergence speed and accuracy when dealing with correlated signals.



(a) (b)

Fig. 7. Time-domain signals of the PEM using IPNLMS l_1-norm algorithm: (a) speech incoming signal, (b) loudspeaker signal.

To provide visual confirmation for these quantitative results, Figure 7 presents a time-domain snapshot of the system operating with the IPNLMS l_1-norm algorithm observed from scope blocks in Simulink. Figure 7(a) shows the original speech incoming signal, while Figure 7(b) displays the corresponding loudspeaker output signal. A direct comparison confirms that the loudspeaker signal closely tracks the waveform and amplitude contours of the incoming signal. The output signal is free from the runaway amplification or distortion characteristic of howling. This observed stability and faithful signal reproduction serve as a practical validation of the numerical metrics. The low $\overline{MIS}$ and high $\overline{ASG}$ of the PEM- IPNLMS l_1-norm algorithm reported in Table 1 are thus shown to directly correspond to an effectively stabilized system.

5.3 Discussion

The interactive learning platform proposed in this work is crucial for clearly demonstrating the fundamental trade-offs inherent in AFC algorithm design. By simultaneously visualizing the MIS curve and monitoring computational load, the platform

makes the abstract trade-off between NLMS and APA tangible. Users can directly see APA's accelerated convergence with speech signals, while also observing the increased processing demands required for its matrix inversion step.

The platform allows users to instantly verify the effectiveness of algorithms designed for sparse impulse responses. By switching from a dense to a sparse feedback path model, users can witness the marked improvement in convergence speed offered by IPNLMS and IPAPA, reinforcing the principle of proportionate adaptation.

The unique objective of H-NLMS is to be made clear through simulation of abrupt system changes. The platform serves as an ideal tool to trigger such events, enabling users to observe how H-NLMS rapidly recovers from instability—a direct visualization of its design to prioritize fail-safe robustness, overachieving the lowest possible steady-state error.

The platform's ability to toggle the PEM-based pre-filtering block provides a direct and powerful method for demonstrating a core theoretical concept in AFC. By enabling or disabling this feature, users can visually and quantitatively observe the consequences of estimation bias. As shown in our experiments, when pre-filtering is disabled, the NLMS algorithm exhibits noticeably slower convergence and a significantly higher steady-state MIS, thus validating the platform as an effective tool for reinforcing established theory in a practical manner.

6 Conclusion and Future Directions

In this work, an interactive learning and research platform has been successfully designed and implemented within the MATLAB/Simulink environment, creating a comprehensive and flexible tool for real-time AFC systems.

The platform addresses a gap in AFC research by providing a unified, interactive environment that allows users to seamlessly switch between various AFC algorithms (such as NLMS, IPNLMS, APA, IPAPA, H-NLMS), toggle pre-filtering blocks, and monitor key performance metrics (MIS and ASG) in real-time. The platform has successfully demonstrated its utility in comparing AFC algorithms and visualizing their real-time performance trade-offs under diverse acoustic scenarios.

To further enhance the platform's capabilities, several future directions are proposed. First, the integration of more advanced algorithms is planned, such as the IPAPSA and its variant or Kalman filter-based approaches. Second, to facilitate practical deployment, development includes automatic code generation for deployment onto real-time hardware targets, such as DSP boards or the Jetson Nano. Lastly, the development of a standalone GUI application will be undertaken to ensure wider accessibility for students and researchers.

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